Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: OWLabs
b. Team leader name: Chirag HS
c. Problem Statement: Manufacturing: Intelligent Operations & Industrial Efficiency

Brief about the idea

TwinForge AI

Your factory. Reimagined before reality.

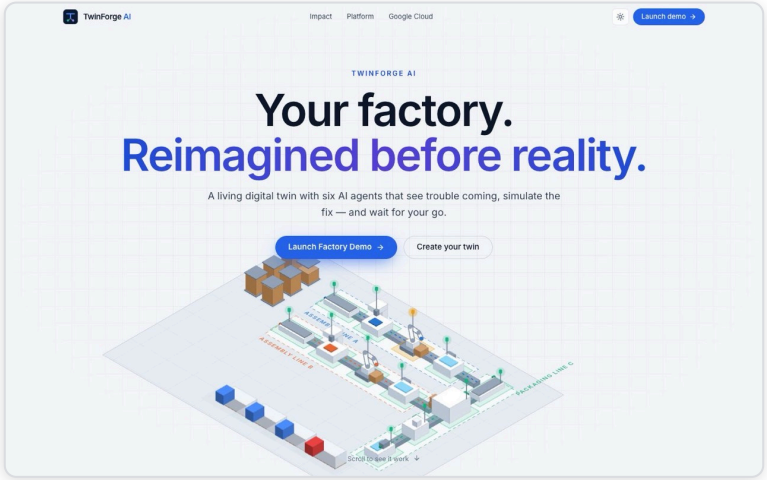
TwinForge AI is a Google Cloud-powered industrial digital twin that connects factory operations, AI agents, and deterministic simulation to help manufacturers predict, optimize, and act with confidence.

Four intelligent capabilities.

- **Predictive Maintenance —** Anticipate equipment failures before they disrupt production.
- **Demand Forecasting —** Predict demand and optimize production plans.
- **Supply Chain Intelligence —** Identify disruptions and optimize materials, inventory, and deliveries.
- **Visual Quality Inspection —** Detect defects, improve first-pass yield, and reduce quality losses.

See. Understand. Simulate. Act.

Explore a live digital twin, investigate incidents with specialist AI agents, simulate alternative scenarios, and execute approved actions only after human authorization.



OUR PRINCIPLE

Language models explain, extract and orchestrate.
Numbers come from configured rules.
Suggestions come from the ontology.
Decisions come from people.

Opportunities

How is it different?

- Not a monitoring dashboard: the twin simulates forward, so options are compared before the plant changes.
- Not a chatbot that guesses: agents compute every figure on a deterministic plant engine and show the evidence.
- Not months of modelling: the Twin Builder drafts a twin from a drawing and a plain-language description.

How will it solve the problem?

- Today, health, quality, materials and orders sit in separate systems, and comparing options by hand takes hours.
- TwinForge joins them in one live twin, predicts trouble early and ranks plans on delivery, cost, risk and sustainability.
- Demo crisis: the recommended plan avoids about 700 late units at about a quarter of the cost of carrying on (model estimates).

USP of the solution

- Models explain, configured rules compute, ontology suggests, people decide: every figure the AI states is checked against configured rules.
- A person approves every plan, with rollback and a full audit trail; there is no control path to equipment.
- Open by design: the same 12 tools are served over MCP, so other agents can use the twin, and still only propose.

List of features offered by the solution

Command Center

OEE, output, machine health, energy, scrap, shortages and delivery risk on one screen.

AI Operations

Predictive Maintenance to anticipate equipment failures, Demand Forecasting to optimize production planning, and Supply Chain Intelligence to identify disruptions.

What-If Simulator

Compare plans on more than ten levers; results recalculate instantly in the browser.

Digital Twin Studio

Topology, isometric, floor-plan and real-time 3D views of one live twin.

Approvals and audit

A person approves, the twin executes, and every step is recorded and reversible.

Visual Quality and CCTV

FPY, DPMO and defect Pareto, nine cameras, and AI visual checks a person confirms.

Twin Builder

From a floor plan and a description to a running twin, with provenance on every element.

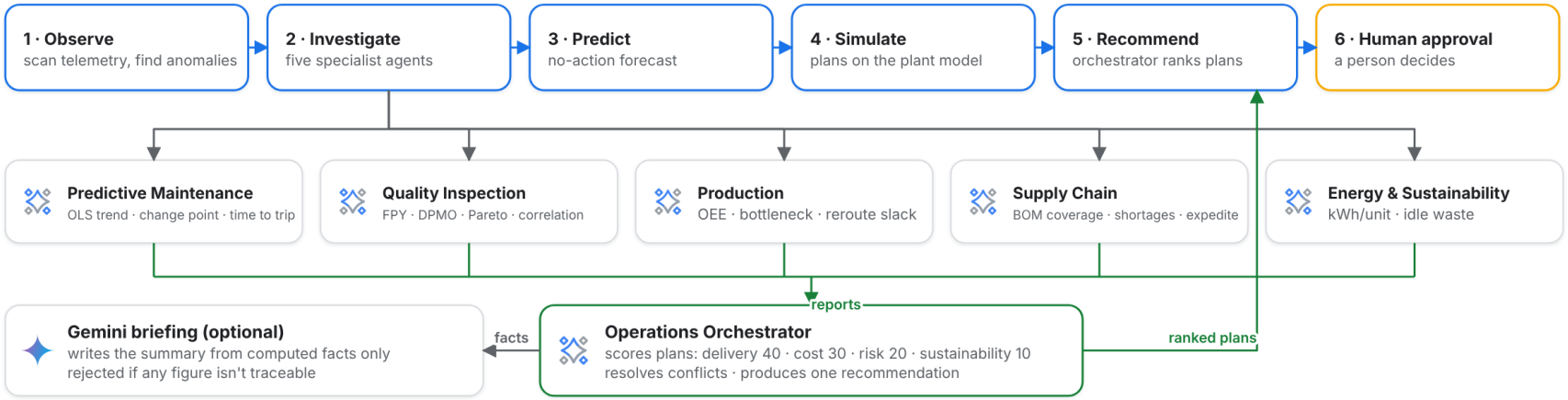
Playground

Drag machines and robots onto the floor, set automation rules, preview in 3D and apply.

Copilot on ADK + MCP

Multi-agent copilot with visible tool calls; the same 12 tools are served over MCP.

Process flow diagram or Use-case diagram



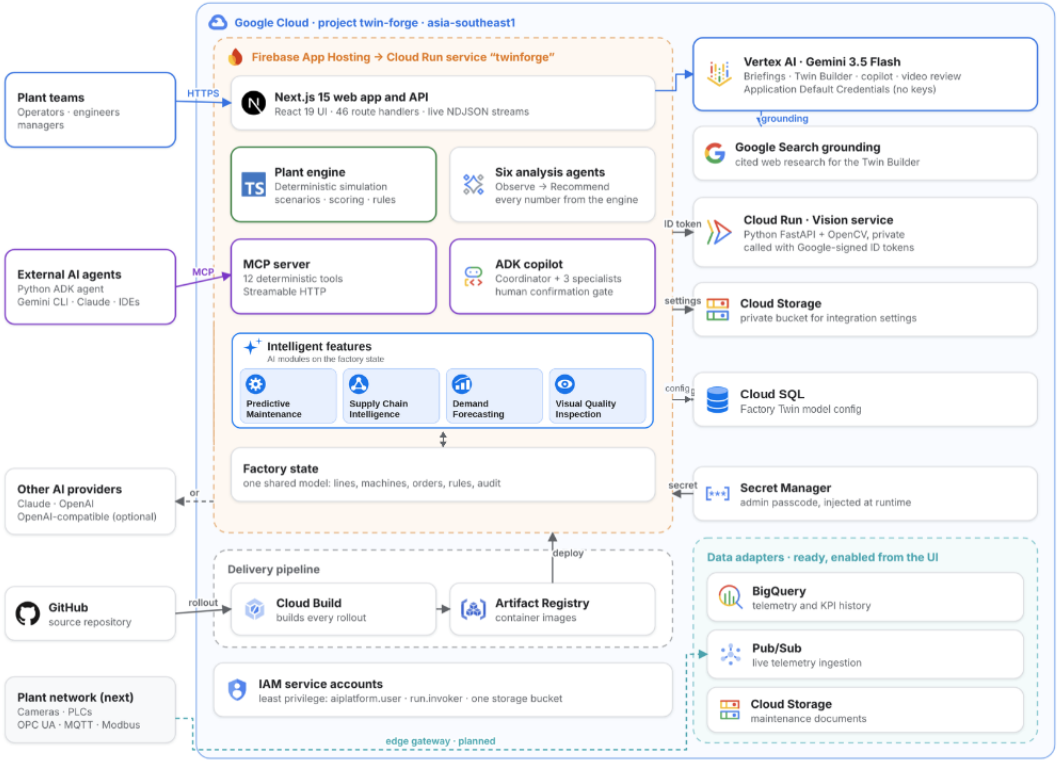
Every finding lists its inputs, method, evidence and confidence. The workflow runs on a frozen snapshot and streams each stage live.
Agents can only propose: the recommendation waits in Approvals until a person approves, rejects or rolls it back.

After approval

Execute the twin runs the plan as directives over simulated time

Audit every step recorded with links; any step can be rolled back

Architecture diagram of the proposed solution



One service

Next.js on Firebase App Hosting (Cloud Run) runs the UI, API, plant engine, ADK copilot and MCP server.

Gemini on Vertex AI

Gemini 3.5 Flash through Application Default Credentials, so no API keys.

Private vision

FastAPI + OpenCV on Cloud Run, called with Google-signed ID tokens.

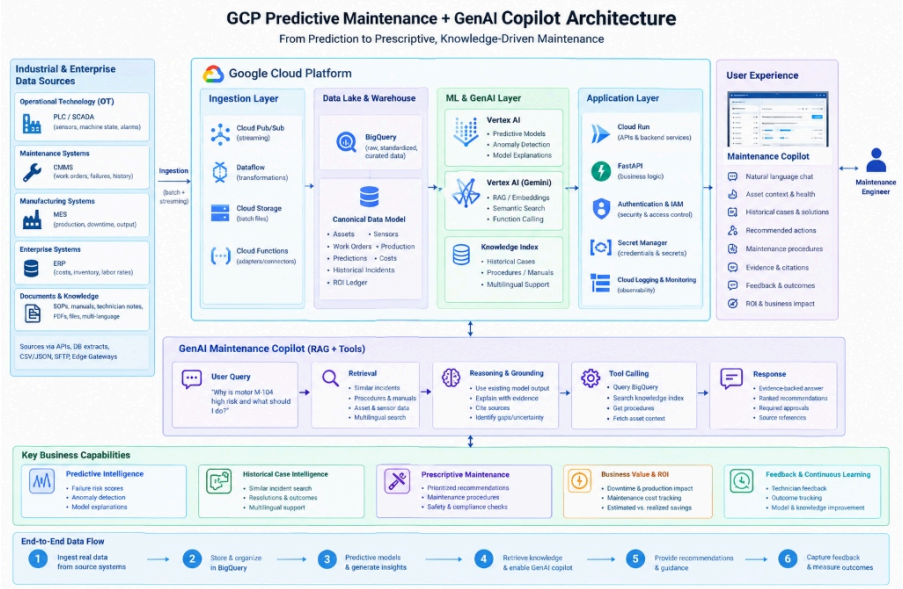
Secure by default

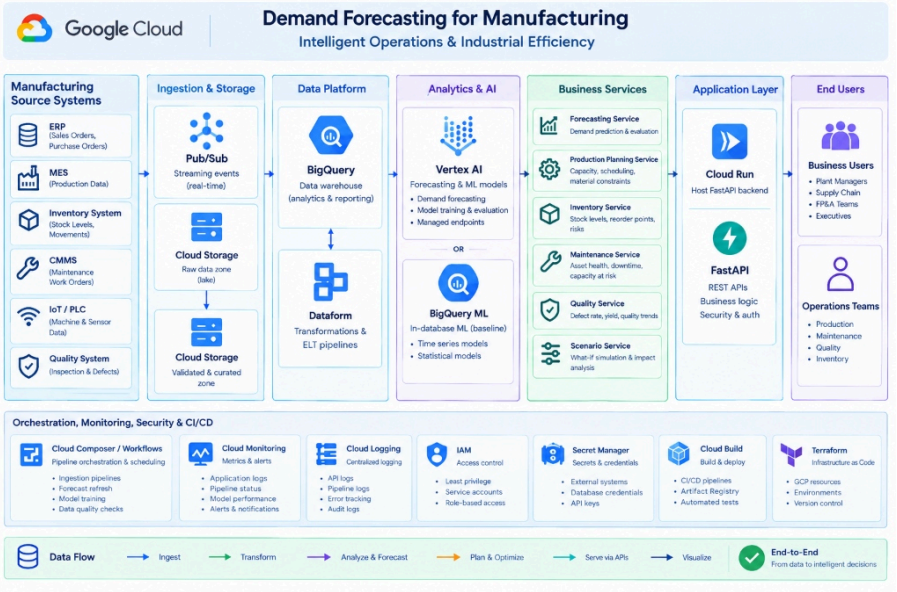
Secret Manager, a private Cloud Storage bucket and least-privilege IAM.

Ready to scale

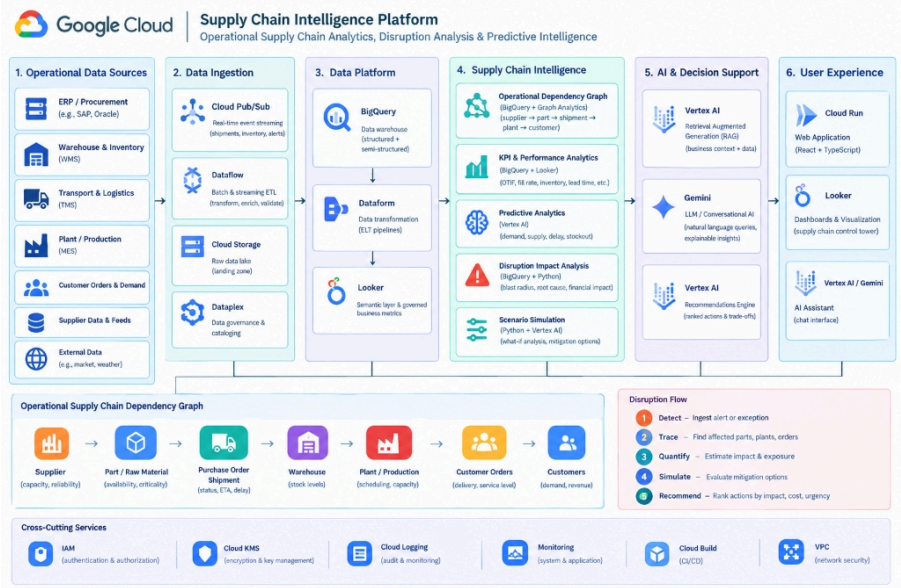
BigQuery and Pub/Sub adapters built in; runs in asia-southeast1.

Architecture diagrams of individual features (1/2)





Architecture diagrams of individual features (2/2)

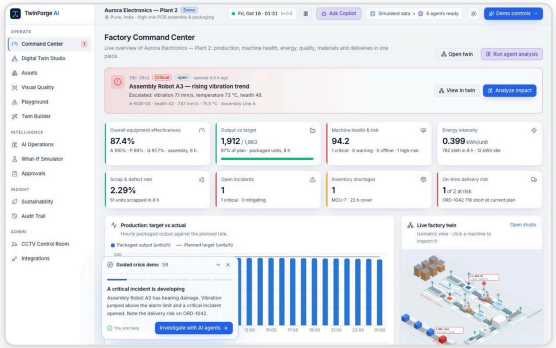




Technologies to be used in the solution

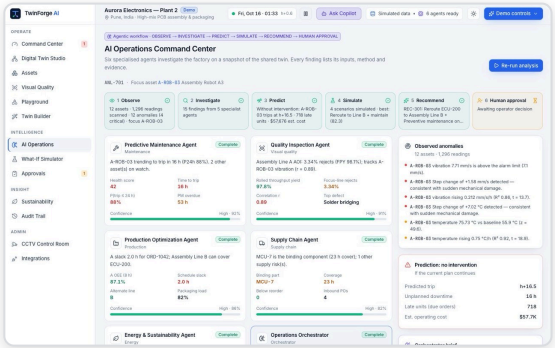
AI and agents	Vertex AI Gemini 3.5 Flash, Google Gen AI SDK, Agent Development Kit TypeScript + Python, Google Search grounding, Model Context Protocol
Google Cloud	Firebase App Hosting, Cloud Run, Cloud Build, Artifact Registry, Secret Manager, Cloud Storage, IAM, BigQuery, Pub/Sub, Cloud SQL
Web application	Next.js, React, TypeScript, Tailwind CSS, Zustand, Zod, React Flow, Recharts, Radix UI, Node.js
3D and media	three.js, React Three Fiber, drei, postprocessing, hls.js
Python services	Python, FastAPI, OpenCV, google-adk[mcp]
Testing	Vitest, Playwright, pytest, GitHub

Snapshots of the prototype (1/2)



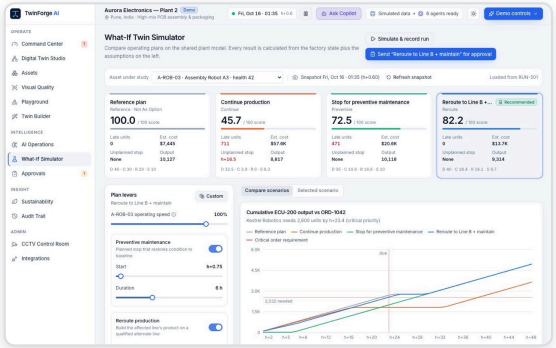
Command Center

The whole plant on one screen during the crisis, with the guided demo panel.



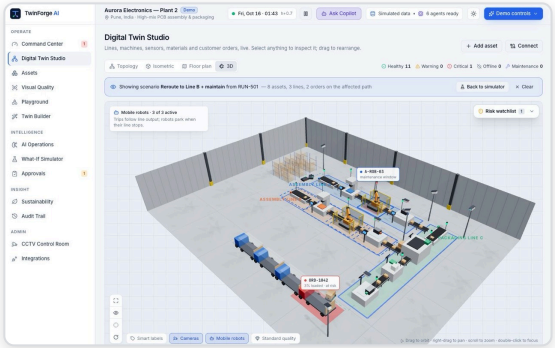
AI Operations

Six agents stream findings; a trip is predicted about 16 hours ahead.



What-If Simulator

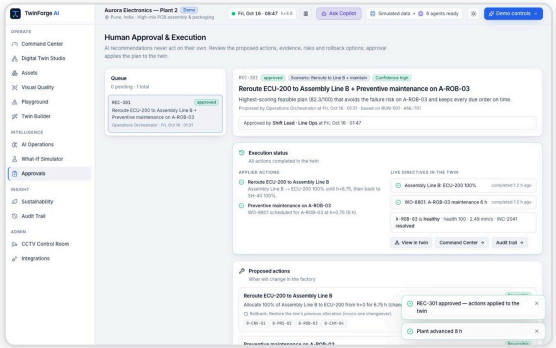
Four plans compared; reroute to Line B + maintain scores 82.2 with no late units.



Digital Twin in 3D

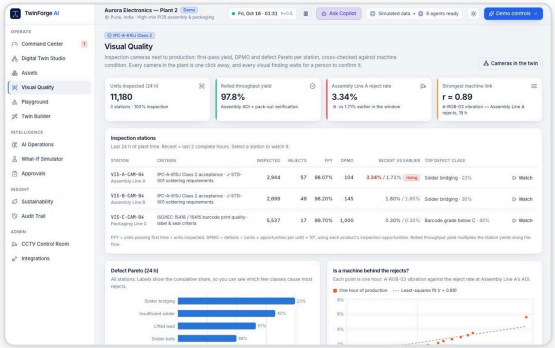
Orbit the live plant; robots animate and the recommended path is highlighted.

Snapshots of the prototype (2/2)



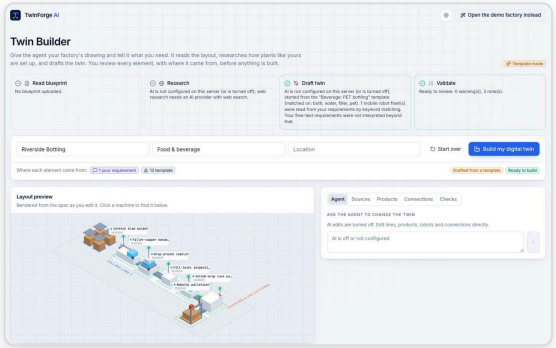
Approvals

The approved plan ran: Line B took over and the failing robot is healthy again.



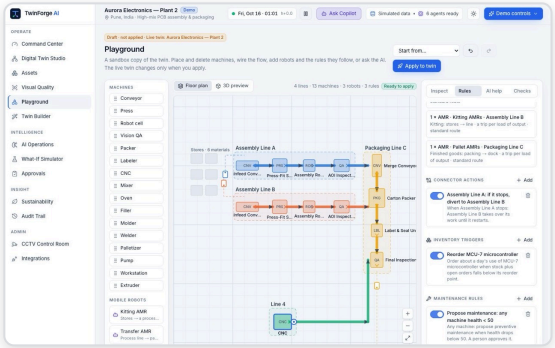
Visual Quality

FPY, DPMO and defect Pareto; robot vibration tracks Line A rejects (r = 0.89).



Twin Builder

Review screen with stages, provenance counts and a live layout preview.



Playground

Place machines and robots on the floor and set automation rules.

Prototype Performance report/Benchmarking

Benchmark: plans compared on the twin in the demo crisis

model estimates on demo data

Plan	Score / 100	Late units	Estimated cost	Unplanned stop
Reference plan (asset assumed healthy)	100.0	0	$7.4K	None
Continue production	45.7	711	$57.6K	Trip at about h+16.5
Stop for preventive maintenance	72.5	471	$20.6K	None
Reroute to Line B + maintain (recommended)	82.2	0	$13.7K	None

Scores weight delivery 40, cost 30, risk 20 and sustainability 10 against the reference plan.

Measured on the live prototype

≈ 5.5 s
Gemini agent briefing

8–23 s
Copilot answer with tool calls

≈ 15 s
External Python ADK agent, 4 tool calls

≈ 78 s
Twin Builder draft with web research

Instant
What-If recalculation, in the browser

19 / 19
Copilot figures traced to tool results

30×
Live plant speed against real time

149
Automated tests: 117 unit, 16 E2E, 16 Python

Additional Details/Future Development (if any)

Current scope

- Runs on a modelled plant with sample data, so results are estimates.
- One factory per deployment; state is held per instance.
- Live telemetry arrives through Pub/Sub today; OPC UA, Modbus, and MQTT require an edge gateway.
- Camera footage in the demo is licensed video from other facilities.
- **Predictive Maintenance:** Demonstrates equipment health monitoring and failure-risk predictions using available sample data.
- **Supply Chain Intelligence:** Demonstrates supply chain visibility, disruption analysis, and recommendations using modelled data.
- **Demand Forecasting:** Generates demand forecasts and production planning recommendations based on sample data.
- **Visual Quality Inspection:** Demonstrates AI-assisted defect detection and quality metrics; inspection results require validation against real production images before operational use.

What comes next

Multi-site

Firestore shared state, Firebase Auth sign-in and several factories per tenant.

Plant connectivity

Edge gateway for OPC UA, Modbus and MQTT into Pub/Sub; camera analytics at the edge.

Learned inspection

Trained inspection models, validated with a measurement-system study before use.

A2A Enterprise and Cloud Scheduler

Cloud Scheduler drives the plant clock, with no instance cap; an Agent-to-Agent (A2A) endpoint enables enterprise agents to communicate between factories.

Mobile app

A companion mobile app for plant teams, with real-time alerts and push notifications, on-the-go monitoring of machines and KPIs.

Provide links to your:



GitHub Public Repository

https://github.com/The-OWLabs/twin-forge-ai



Demo Video Link (3 Minutes)

https://drive.google.com/drive/folders/1_74Eehjv1zZRxFji0yO3JNsuIztxNazU?usp=share_link



Final Product Link

https://twinforge--twin-forge.asia-southeast1.hosted.app

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

